MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, DISPLAY POWER SUPPLY

This specification is approved for use by the US Army Armament, Munitions and Chemical C-and and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification establishes the requirements for manufacture and acceptance of the Display Power Supply Circuit Card Assembly: 9355847, used in the Mortar Ballistics Computer (MBC): M23. Throughout this specification this assembly is referenced as "supply".

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standasrds of the issue listed in that issue of the Department-of Defense index of Specifications and Standards (DoDISS) Specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel; General Specification Governing the Manufacture and Inspection
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-454	Electronic Equipment, General Requirements for
MIL-STD-1521	Technical Reviews and Audits for Systems, Equipments, and Computer Programs
MIL-STB45662	Calibration System Requirements

Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Goverment documents, drawings, and publications. the following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect in date of invitation for bids or request for proposal.

DRAWINGS

9355847	Circuit Card Assembly, Display Power Supply
11785850	Computer Set, Ballistics, Mortar: M23

PACKAGING DATA SHEETS

9355847 -	Packaging of Circuit Card Assembly, Power Supply, Display: 9355047

Copies of drawings and packaging data “sheets required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Item definition. The display power supply circuit card assemblv shall Provide the Mortar Ballistic Computer (MBC) with display-and electroluminescent drive voltages from various battery voltages or from unregulated external prime power. It also supplies control logic for the MBC logic power supply and a RAM standby battery with charging circuitry. The connector contacts are listed in Table I.

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TABLE I. Display power Supply circuit card assembly connector contacts.

Connector	Pin	Type	Name
P1	Al	o	+35V
P1	A2	o	ELHI
P1	A3	o	ELLO
P1	A4	I	MC1200
PI	B1	I	ELON
P1	B2	I	DSPON*
Pl	B3	I	ON*
P1	B4	I	OFF
P2	Al	I	ps+5v
P2	A2	I	+5V
P2	A3	I	SPOS
P2	A4	o	PSRUN*
P2	B1	o	BAT+5V
P2	B2		GND
P2	B3		GND
P2	B4	I	PSFAUIJT*

3.2 Performance.

3.2.1 General. The performance of the supply shall be in accordance with following requirements.

3.2.1.1 Display on. The supply shall turn on when P1-B2 is connected to P2-B2 {GDD) and shall turn off when p1-B2 is connected to P2-A1. It shall be considered on when requirement 3.2.1.2 is met.

3.2.1.2 Positive volts. P1-A1 shall be +35 plus or minus 2. volts DC when referenced to -.

3.2.1.3 Battery volts. P2-BI shall be plus S.4 plus or Xninu 0.6 volts DC when referenced to GND.

3.2.1.4 Output ripple. The +35V output shall have an AC cmponent with a voltage level less than 0.6 volts peak to peak that is at a frequency less than 1 megahertz.

3.2.1.5 Overload/shortcircuit protection. The +35V output shall begin to decrease in voltage when the output current is at least 22 milliamperes and at most 55 milliamperes. When the +3%v output is shortcircuited to GND, the output current shall be less than .5 amperes.

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3.2.16 Power supply run. The voltage between P2-A4 and GND hall be .5 plus or minus .5 volts DC when P1-B3 is connected to AND. After P1-B3 is disconnected from GND and then P1-B4 is connected to P2-Al, the voltage between P2-B4 and GND shall be 21 plus or minus 11 volts DC.

3.2.1.7 Electroluminescent panel high. P1-A2 shall be 35 plus Or minus 15 volts rms when referenced to P1-A3 after P1-B1 iS connected to P2-A1 shall be less than 3 volts rxns after P1-B1 as disconnected.

3.2.1.8 Common connector contact continuity. The resistance between GND and P2-B3 shall be less than 7.5 ohms.

3.2.2 Physical characteristics. The physical characteristics shall be as prescribed by the drawings,specifications, and QAP's.

3.2.3 Reliability. When specified in the contract, a special sample shall be subjected to reliability testing. The manufacturing and production techniques employed in the fabrication of the supply shall not degrade the inherent reliability. The inherent reliability shall not be less than 131,068 hours Mean Time Between Failures (MTBF). (See 4 and 6)

3.3 Environmental. The supply shall meet the requirements specified herein unless otherwise specified.

3.3.1 Room ambient temperature. The requirements of this specification shall be met with the supply thermally stabilized at room ambient temperature of +16°C to +32°C (+g\6O°F to +90°F) , unless otherwise specified in the requirements paraqraph.

3.3.2 Environmental stress screen (ESS) temperature. The supply shall meet the requirements of 3.2.1.1 and 3.2.1.8 inclusive, after having been exposed and to a minimum of ten temperature cycles at temperatures of -57°C and +71°C (-71°F and +160°F).

3.3.3 Environmental stress screen (ESS) vibration. The supply shall meet the requirements of 3.2.1.1 and 3.2.1.8 inclusive, after having been subjected to a random and pseudorandom vibration which includes frequency content from 100 Hz to 1000 Hz and up to a 3g W in two axis for a period of 10 minutes or minus 15 seconds.

3.4 Fabrication. The supply shall be manufactured in accordance with drawing 9355B47.

3.5 First article. When specified in the contract, a sample shall be subjected to first article inspection (See 4 and 6).

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3.6 Workmanship. The workmanship requirements of MIL-F-13926 and MIL-STD-454 shall be met. All components shall be fitted properly and firmly in their respective positions. All electrica: connections shall be mechanically secure and electrically sound. There shall be no evidence of corrosion, burrs, roughness, sharp edges or other defects which could adversely affect its use or suitability for the purpose intended.

3.7 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provosions. The component and subassembly inspection requirements of MIL-F-13926 and MIL-ST-454 form a part of the Quality Assurance provisions of this specification Definitions of inspection terms shall be as listed in MIL-STD-109

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval test shall consist of three (3) supplies plus three (3) each Of all Items covered by QAP. The three supplies shall be tested as specified herein for all the requirements of this specification. The items covered by QAP shall be tested, as specified thereon, for all characteristics delineated on the QAP. The samples shall be manufactured in the same manner, using the Same materials equipment, Processes, and procedures as will be used in the production of the balance of the contract quantity. All parts an materials, including packaging and packing, shall be obtained from the same source of supply as will be used in the production of tthe balance of the contract quantity.

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4.2.1 Government testing. When the Government is responsible or conducting first or article approval tests, the contractor prior submitting the sample to the Government, shall inspect the ample to insure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4.2.2 Contractor testing. men the contractor is responsible or conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for all materials, shall be submitted or the Government for approval. The Government reserves the right so witness any. or all of the contractor's inspections.

4.3 Inspectictions. Provisions

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with Submission of product" provision of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Quality Assurance Provisions (QAP's) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.4 Sampling.

4.4.1 Environmental/performance. The supply shall be inspected to the requirements in Table II.

TABLE II. Classification of defects.

No. 100% Inspection	Characteristics	Requirement	Test Procedure
1.	Display on	3.2.1.1	4.6.3.1
2.	Positive volts	3.2.1.2	4.6.3.2
3.	Battery volts	3.2.1.3	4.6.3.3
4.	Output ripple	3.2.1.4	4.6.3.4
5.	Overload/shortcircuit protection	3.2.1.5	4.6.3.5
6.	Power supply run	3.2.1.6	4..6.3.6
7.	Electroluminescent panel high	3.2.1.7	4.6.3.7

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8.	Common connector contact continuity	3.2.1.8	4.6.4
9.	ESS temperature	3.3.2	4.6.1
10.	ESS vibration	3.3.3	4.6.2
MAJOR: AQL 1.0% Defective			
101.	Workmanship	3.6	4.6.5

4.4.2 Special sampling. One supply shall be selected at random as a special sample from each lot or each 100 produced which ever occurs first. The sample shall meet the requirements and tests in Table 111.

TABLE 111. Classification of defects.

No.	Characteristics	Requirement	Test Procedure
201.	Fabrication	3.4	Applicable Drawing/ Visual
202.	Materials	3.7	Applicable Drawing/ Visual

Note: The tests in Table III shall be conducted at room ambient temperature (as specified in 3.3.1).

4.4.3 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment. Supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. All inspection equipment specified by drawing number In specification or QAP forming a pa of the contract shall be supplied by the contractor in accordance with technical data included in the Technical Data Package List (TDPL).

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4.5.2.2 Contractor design. In the absence of Government inspection equipment design, the contractor shall design and supply Inspection equipment compatible with the “Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and ‘Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error In the test equipment design must be considered part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Temperature testing. The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F (1.4°C). The temperature gradient the cross-sectional area occupied by the test item shall not exceed 0.5°F per foot in any direction. The heat source of the test chamber shall be so located that radient heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber within the supply airstream, or in the return airstream, whichever provides the 6pecified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.5.2.3.2 Fabrication, setup and calibration,Design and fabrication of test equipment shall be such that it incorporates the features of rigidity~ stability, and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment. Operating and calibration instruction shall be prepared and submitted to the procuring activity.

4.6 Test metdods and procedures. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the government procuring agency for approval prior to use.

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4.6.1 ESS temperature. Testing equipment utilized in this test shall conform to the requirements of 4.5.2.3.1. The supply shall be placed in the test chamber and the temperature of the chamber reduced 10°C (18°F) per minute plus or minus 1°C (1.8°F) from room ambient temperature (as specified in 3.3.1) to -57°C (-71°F). The supply shall be maintained at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure, the temperature of the chamber shall be raised 10°C (18°F) per minute plus or minus 1°C (1.8°F) to +71°F (+160°C). The supply shall remain at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure the temperature of the chamber shall be reduced 10°C (18°F) per minute plus or minus 1°C (1.8°F) to room ambient temperature (as specified in 3.3.1). The above procedure shall be conducted a minimum of ten times. Upon completion of the cycles the supplY Shall then be removed from the chamber and subjected to the tests in 4.6.3.1 and 4.6.4 inclusive. It shall meet the requirements of 3.2.1.1 and 3.2.1.8 inclusive.

4.6.2 ESS vibration. Conduct the vibration test with the supply positioned on an appropriate adapter affixed to a vibrating machine that is capable of providing the amplitudes and frequencies specified in 3.3.3. Unless otherwise specified, the test equipment shall be in accordance with the “Test Facilities” requirements of MIL-F-13926. The applied motion shall be in two separate planes for the time periti specified. It shall meet the requirements of 3.2.1.1 and 3.2.1.8 inclusive. Upon completion of the vibration sweep the supply shall be removed from the vibration adapter and shall be subjected to the tests in 4.6.3.1 and 4.6.4 inclusive. It shall meet the requirements of 3.2.1.1 and 3.2.1.8 inclusive.

4.6.3 Performance. The performance tests 4.6.3.1 and 4.6.3. inclusive, shall be conducted with a 100 kilohn plus or minus 5 percent 118 watt resistor between P2-A2 and P2-B4, a second between P2-A4 and P2-A3, and a third between P1-B2 and P2-82; a 5.3 plus or minus .1 volts DC power supply to P2-A1 and P2-A2 bot with respect to GND; a 1200 plus or minus 12 Hz square wave generator that has a high level of 4.6 plus or minus .7 volts DC and a low level of .5 plus or minus .5 volts DC to P1-A4 with respect to GND; a 43.2 kilohm plus or minus 1 percent 1/10 watt resistor in parallel with a .047 microfarad plus or minus 1 percent 100 volt AC capacitor. Connect the parallel combination between P1-A2 and P1-A3; and a 21 plus or minus 11 volts DC power supply to P2-A3 with respect to P2-B2.

4.6.3.1 Display on.

4.6.3.2 Positive volts. Turn on the supply while monitoring P1-A1 with a voltmeter. The output response shall be as specified in 3.2.1.2.

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4.6.3.3 Battery volts. After P2-A3 has been connected to a voltage source as specified in 4.6.3, for 30 to 45 minutes disconnect the voltage. Measure the voltage at P2-B1 with respect to GND. The measured voltage shall be as specified in 3.2.1.3.

4.6.3.4 Output ripple. Turn on the supply. With an oscilloscope, look at the output ripple at the ”+35V output with respect to GND. The output response shall be as specified in 3.2.1.4.

4.6.3.5 Overload/shortcircuit Protection. Conduct this test by connecting a variable (0 to 10 kilohm) 10 watt resistor in series with a 500 milliamp DC current meter. Connect the variable resistor current meter combination between +35V and GND. Set the resistor to its maximum resistance and turn on the supply. slowly decrease the resistance while at the same time noting the current meter reading and the DC voltage level at +35V with respect to GND. The output response shall be as specified in 3.2.1.5.

4.6.3.6 Power supply run. Conduct this test by connecting P1-B3 to GND while monitoring P2-A4 with a voltmeter. Then disconnect P1-B3 from GND and connect P1-B4 to +5-V. The out response shall be as specified in 3.2-1.6.

4.6.3.7 Electrolminescent panel high. Turn on the supply while monitoring the AC voltage between P1-A2 and P1-A3. Connect P1-B1 to +5V. The output response shall be as specified in 3.2.1.7.

4.6.4 Common connector contact continuity. This test shall be conducted with no external power connected to the supply. Measure with an ohm meter the resistance between GND and P2-B3. The resistance shall be as specified in 3.2.1.8.

4.6.5 Workmanship. The supply shall be subjected to a visual and tactile examination for evidence of faulty workmanship. The supply shall conform to the requirements of 3.6.

4.7 Reliability testing. The requirement for reliabilty test approval and the responsibility (Government or contractor) for reliability testing shall be as-specified in the contract. The sample for reliability test shall consist of 3 supplies. The samples shall be manufactured in the same manner, using the saw materials equipment, processes, and procedures as will be used in the production of the balance of the contract quantity. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

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4.7.1 Government testing. When the Government is responsible for conducting reliability tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to assure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4.7.2 Contractor testing. When the contractor is responsible for conducting reliability tests, the sample shall be inspeced and tested by the contractor for all the requirements the contract. The sample and a record of these inspections and tests, including certificates of conformance for all materials, shall be submitted to the Government for approval. The Government, reserves the right to witness any or all of the contractors reliability tests and inspections.

5. PREPARATION FOR DELIVERY

5.1 Delivery requirements. Handling, packaging, packing, shipping, and marking of the Display Power Supply Logic Circuit Card Assembly, 9355847 shall be in accordance with Packaging Data Sheet 9355847. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The display described herein is intended for use in the MBC. See drawing 11785850.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for first article testing.

c. Provisions for reliability testing.

d. Applicable Packaging Data Sheet.

e. Applicable levels of preservation, packing, and marking (see 5.1).

f. Applicable stock numbers.

Custodian:
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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70495 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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